HW4

2025-03-10

เอาHW 3 มาเทียบ (ตัว HW 4 อยู่บรรทัดที่ 349)

title: “HW3” output: word_document: default html_document: default date: “2025-03-08” — ### Data: BankChurners #### Variables: #### “Attrition_Flag” #### “Customer_Age” #### “Gender” #### “Dependent_count” #### “Education_Level” #### “Income_Category” #### “Months_on_book” #### “Total_Relationship_Count” #### “Months_Inactive_12_mon” #### “Contacts_Count_12_mon” #### “Credit_Limit” #### “Avg_Utilization_Ratio”

แปลงค่า Non-Numeric

```
dim(bank)
```

```
## [1] 10127    24
```

```
table(bank$Attrition_Flag)
```

```
##
## Attrited Customer Existing Customer
##              1627              8500
```

Smote 75% (ไม่ 100% เพราะ ป้องกัน Overfitting)

```
library(performanceEstimation)
bank <- smote(Attrition_Flag~., data = bank, perc.over = 3.45, perc.under = 1.75)

dim(bank)
```

```
## [1] 15049    24
```

```
table(bank$Attrition_Flag)
```

```
##
## Attrited Customer Existing Customer
##              6508              8541
```

#Model 1 ##Logistic Regession with Monte Cross-Validation

```
avg_table
```

```
##                          Attrited Customer Existing Customer
## Pred Attrited Customer             976.0            529.05
## Pred Existing Customer             325.4           1179.55

avg_acc

## [1] 0.7161296

avg_pre

## [1] 0.6484831

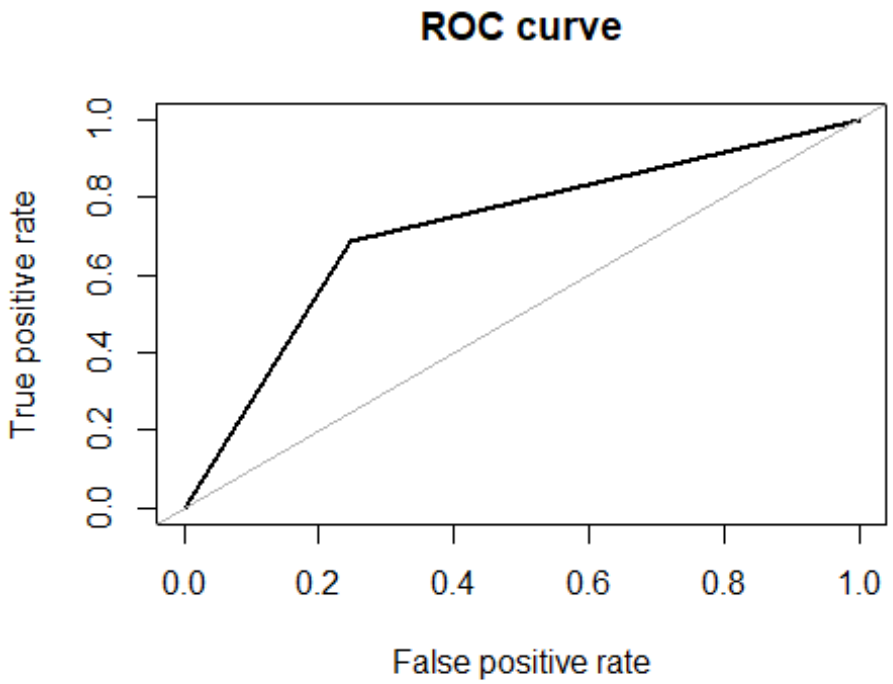
avg_rec

## [1] 0.7499732

avg_f1

## [1] 0.695468

library(ROSE)
roc.curve(bank$Attrition_Flag[-train], pred)
```




```
## Area under the curve (AUC): 0.721
```

Model 2

เทียบ forward backward hybrid จาก rss

จำนวนตัวแปรที่ให้ ค่า rss น้อยสุด ของแต่ละ method

```
varia_select
```

```
##      Method     number of variable
## [1,] "Forward"  "23"
## [2,] "Backward" "23"
## [3,] "Hybrid"   "23"
```

เลือกใช้ hybrid method เพื่อความหยืดหยุ่นกับชุดข้อมูล

```
avg_hyb_table
```

```
##                         Attrited Customer Existing Customer
## Pred Attrited Customer            976.45            528.65
## Pred Existing Customer            323.80           1181.10
```

```
avg_hyb_acc
```

```
## [1] 0.716794
```

```
avg_hyb_pre
```

```
## [1] 0.6487609
```

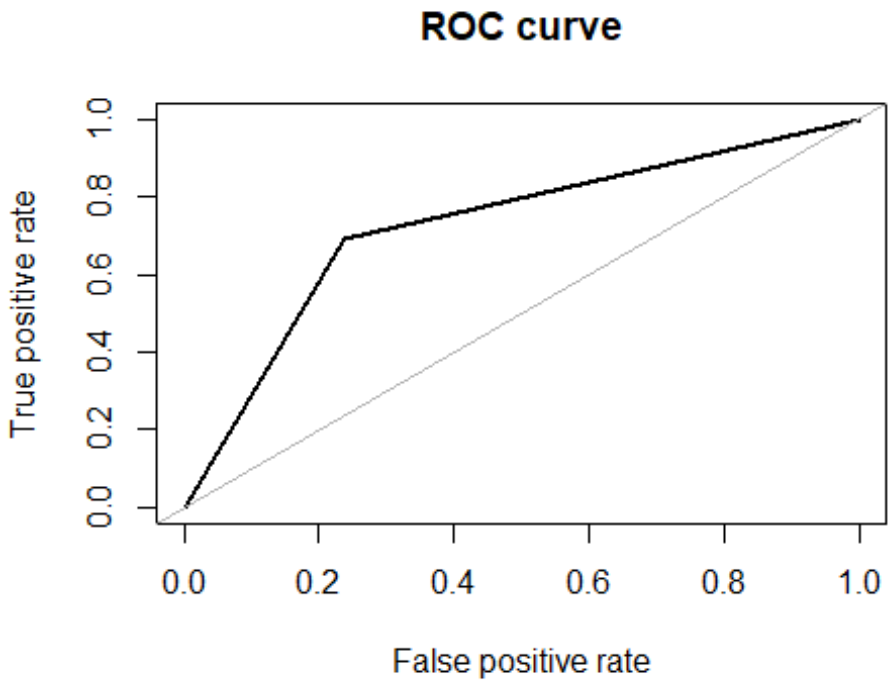
```
avg_hyb_rec
```

```
## [1] 0.7509974
```

```
avg_hyb_f1
```

```
## [1] 0.6961094
```

```
roc.curve(bank[-train,]$Attrition_Flag, hyb_pred)
```



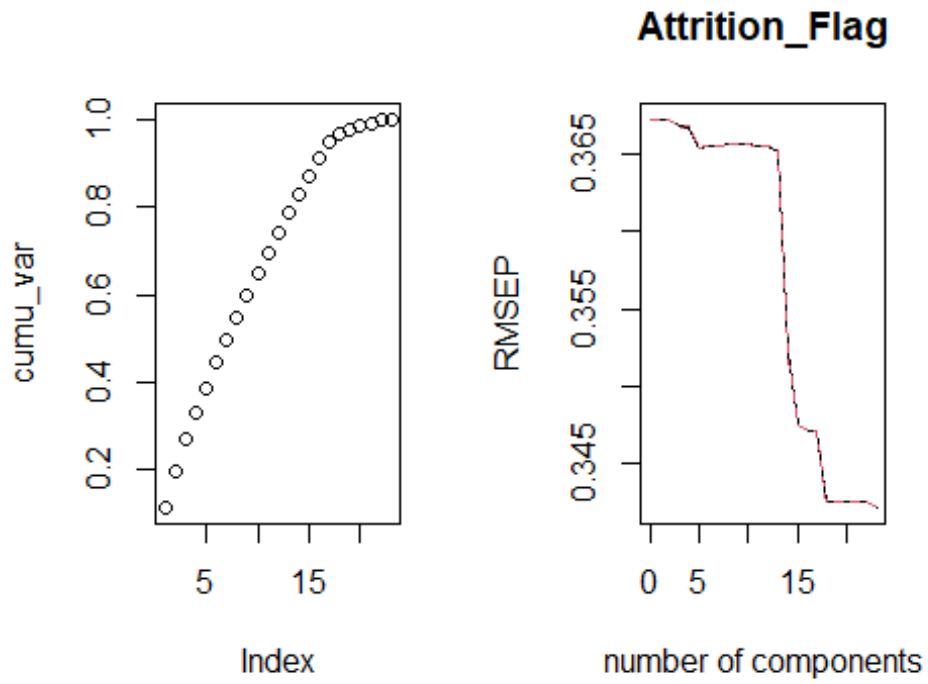

```
## Area under the curve (AUC): 0.727
```

Model 3

PCA

compare cumulative with cross-validation error

```
par(mfrow = c(1,2))
plot(cumu_var)
validationplot(pca_mod, val.type="RMSEP")
```




```
cumu_var[18]
```

```
## [1] 0.9669408
```

จากการพิจารณากราฟ จึงเลือก PCA 18

```
bank_pca <- data.frame(bank$Attrition_Flag, pca$x[,1:18])
bank_pca$bank.Attrition_Flag <- factor(bank_pca$bank.Attrition_Flag, levels =
c(0,1))

library(performanceEstimation)
bank_pca <- smote(bank.Attrition_Flag~., data = bank_pca, perc.over = 3.45,
perc.under = 1.75)

pca_mod <- glm(bank.Attrition_Flag~.,data = bank_pca, family = "binomial")

avg_pca_table
```

```
##                          Attrited Customer Existing Customer
## Pred Attrited Customer             965.15            539.85
## Pred Existing Customer             331.05           1173.95

avg_pca_acc

## [1] 0.7106645

avg_pca_pre
```

```
## [1] 0.6412957
```

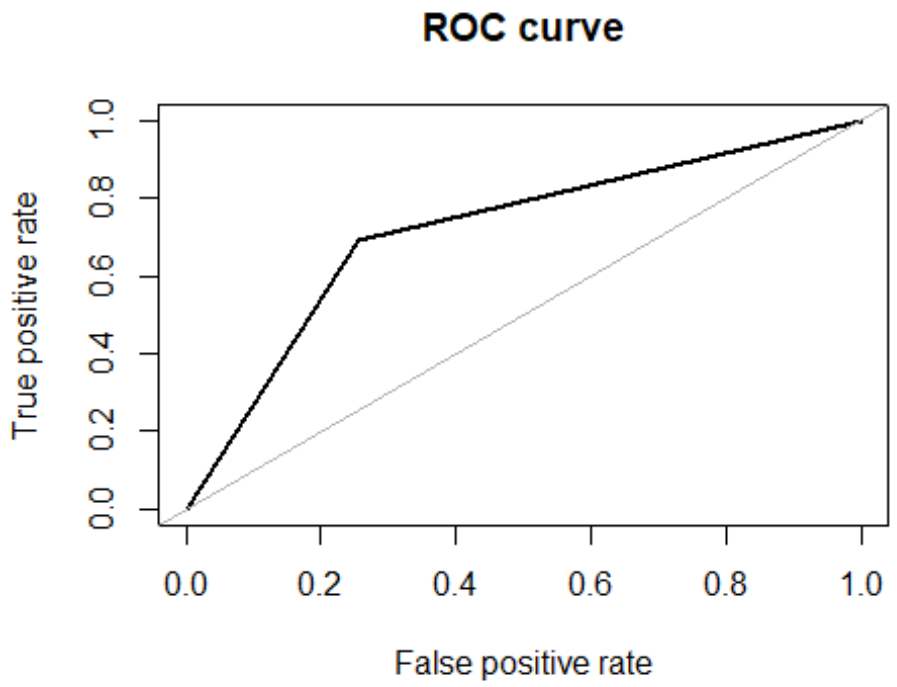
```
avg_pca_rec
```

```
## [1] 0.7446498
```

```
avg_pca_f1
```

```
## [1] 0.6890582
```

```
roc.curve(bank_pca[-train,]$bank.Attrition_Flag, pca_pred)$auc
```




```
## [1] 0.7170075
```

เปรียบเทียบ 3 model

```
compare_table
```

```
##             Model_1   Model_2   Model_3
## accuracy  0.7161296 0.7167940 0.7106645
## precision 0.6484831 0.6487609 0.6412957
## recall    0.7499732 0.7509974 0.7446498
## f1-score  0.6954680 0.6961094 0.6890582
## AUC       0.7211240 0.7268938 0.7170075
```

จากการที่ จุดประสงค์ในการสร้างตัวแบบ คือการ “หาลูกค้าที่ยกเลิกบัตร”
และจำเป็นต้องคำนึงถึงลูกค้าที่ยังคงเป็นลูกค้าบัตรอยู่ จึงใช้ AUC ทดสอบ ภาพรวมของ Model

จากการที่ Model_2 ให้ค่า AUC ที่มากที่สุด จึงจะนำ MOdel_2 มาพิจารณา

```
summary(hyb_mod)
```

```
##
## Call:
## glm(formula = Attrition_Flag ~ ., family = "binomial", data = bank[,
##     hyb_variable], subset = train)
##
## Coefficients:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                    3.636e-01  2.421e-01   1.502 0.133060
## Customer_Age                  -1.207e-02  4.598e-03  -2.625 0.008661 **
## GenderM                        6.125e-01  8.730e-02   7.016 2.28e-12 ***
## Dependent_count               -7.855e-02  1.778e-02  -4.417 1.00e-05 ***
## Education_LevelDoctorate      -3.311e-01  1.177e-01  -2.814 0.004898 **
## Education_LevelGraduate       -5.715e-02  7.611e-02  -0.751 0.452680
## Education_LevelHigh.School     5.360e-02  8.128e-02   0.660 0.509567
## Education_LevelPost.Graduate  -4.318e-01  1.140e-01  -3.787 0.000153 ***
## Education_LevelUneducated     -5.297e-02  8.579e-02  -0.617 0.536926
## Education_LevelUnknown        -7.760e-02  8.587e-02  -0.904 0.366115
## Marital_StatusMarried          7.582e-02  8.888e-02   0.853 0.393637
## Marital_StatusSingle          -4.175e-02  8.956e-02  -0.466 0.641090
## Marital_StatusUnknown         -1.116e-02  1.159e-01  -0.096 0.923283
## Income_Category.40K....60K     9.001e-01  1.142e-01   7.880 3.26e-15 ***
## Income_Category.60K....80K     6.357e-01  1.006e-01   6.320 2.61e-10 ***
## Income_Category.80K....120K    3.175e-01  9.537e-02   3.330 0.000870 ***
## Income_CategoryLess.than..40K  7.726e-01  1.266e-01   6.100 1.06e-09 ***
## Income_CategoryUnknown         1.015e+00  1.343e-01   7.555 4.18e-14 ***
## Months_on_book                 1.270e-02  4.614e-03   2.753 0.005904 **
## Total_Relationship_Count       3.391e-01  1.448e-02  23.427  < 2e-16 ***
## Months_Inactive_12_mon        -5.165e-01  2.411e-02 -21.423  < 2e-16 ***
## Contacts_Count_12_mon         -6.273e-01  2.241e-02 -27.990  < 2e-16 ***
## Credit_Limit                   4.638e-05  3.157e-06  14.691  < 2e-16 ***
## Avg_Utilization_Ratio          2.783e+00  9.435e-02  29.498  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 16483  on 12038  degrees of freedom
## Residual deviance: 13106  on 12015  degrees of freedom
## AIC: 13154
##
## Number of Fisher Scoring iterations: 4
```

HW4

Lasso Regression

```
bank <- read.csv("BankChurners.csv")

bank <- bank[,c(2:8, 10:14, 21)]
bank$Attrition_Flag <- factor(bank$Attrition_Flag, levels = c("Attrited
Customer", "Existing Customer"))

table(bank$Attrition_Flag)

##
## Attrited Customer Existing Customer
##              1627              8500

str(bank)

## 'data.frame':    10127 obs. of  13 variables:
##  $ Attrition_Flag          : Factor w/ 2 levels "Attrited Customer",..: 2
2 2 2 2 2 2 2 2 2 ...
##  $ Customer_Age            : int  45 49 51 40 40 44 51 32 37 48 ...
##  $ Gender                  : chr  "M" "F" "M" "F" ...
##  $ Dependent_count         : int  3 5 3 4 3 2 4 0 3 2 ...
##  $ Education_Level         : chr  "High School" "Graduate" "Graduate"
"High School" ...
##  $ Marital_Status          : chr  "Married" "Single" "Married" "Unknown"
...
##  $ Income_Category         : chr  "$60K - $80K" "Less than $40K" "$80K -
$120K" "Less than $40K" ...
##  $ Months_on_book          : int  39 44 36 34 21 36 46 27 36 36 ...
##  $ Total_Relationship_Count: int  5 6 4 3 5 3 6 2 5 6 ...
##  $ Months_Inactive_12_mon  : int  1 1 1 4 1 1 1 2 2 3 ...
##  $ Contacts_Count_12_mon   : int  3 2 0 1 0 2 3 2 0 3 ...
##  $ Credit_Limit            : num  12691 8256 3418 3313 4716 ...
##  $ Avg_Utilization_Ratio   : num  0.061 0.105 0 0.76 0 0.311 0.066 0.048
0.113 0.144 ...

Attrition_Flag <- bank$Attrition_Flag
bank <- model.matrix(Attrition_Flag~., data = bank)[,-1]
bank <- data.frame(Attrition_Flag, bank)

library(performanceEstimation)
#bank <- smote(Attrition_Flag~., data = bank, perc.over = 4.7, perc.under =
1.3)
bank <- smote(Attrition_Flag~., data = bank, perc.over = 3.45, perc.under =
1.75)



library(glmnet)
```

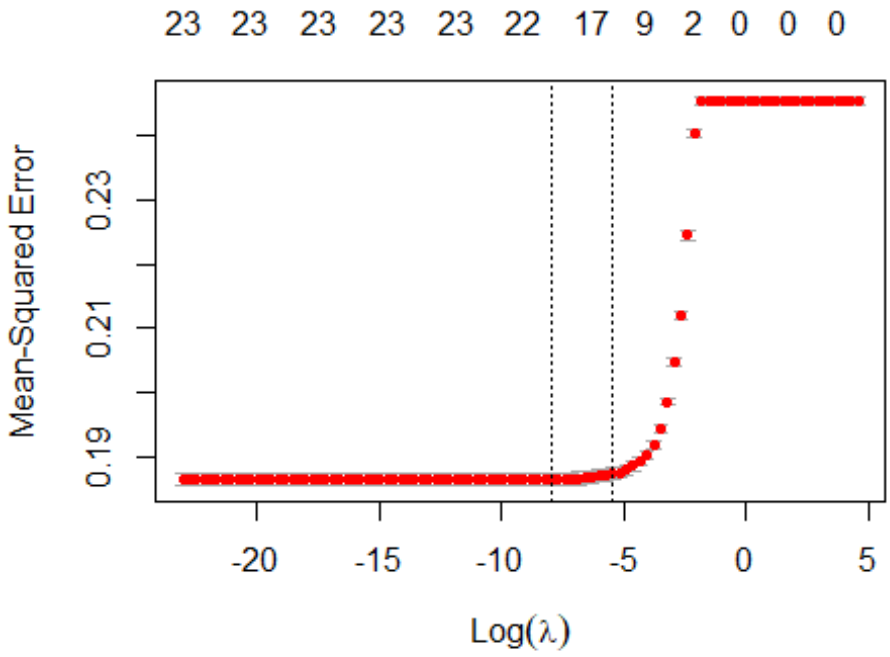
```
## Loading required package: Matrix

## Loaded glmnet 4.1-8

#grid=10^seq(10,-2, length=100)
grid=0.1^seq(10,-2, length=100)

y <- as.numeric(bank$Attrition_Flag)-1
x <- bank[,-1]
x <- scale(x)

cv_lass <- cv.glmnet(x,y,nfolds = 5, alpha=1, lambda = grid)
plot(cv_lass)
```




```
bestlamda <- cv_lass$lambda.min
bestlamda

## [1] 0.0003511192

lass_acc = seq(20)
lass_tp = seq(20)
lass_tn = seq(20)
lass_fp = seq(20)
lass_fn = seq(20)
lass_pre = seq(20)
lass_rec = seq(20)
lass_f1 = seq(20)
```

```r
for (i in 1:20){
  train <- sample(dim(bank)[1], size = (0.8*dim(bank)[1]), replace = FALSE)
  y_train <- as.numeric(bank[train,]$Attrition_Flag)-1
  x_train <- bank[train,-1]
  x_train <- scale(x_train)
  y_test <- as.numeric(bank[-train,]$Attrition_Flag)-1
  x_test <- bank[-train,-1]
  x_test <- scale(x_test)

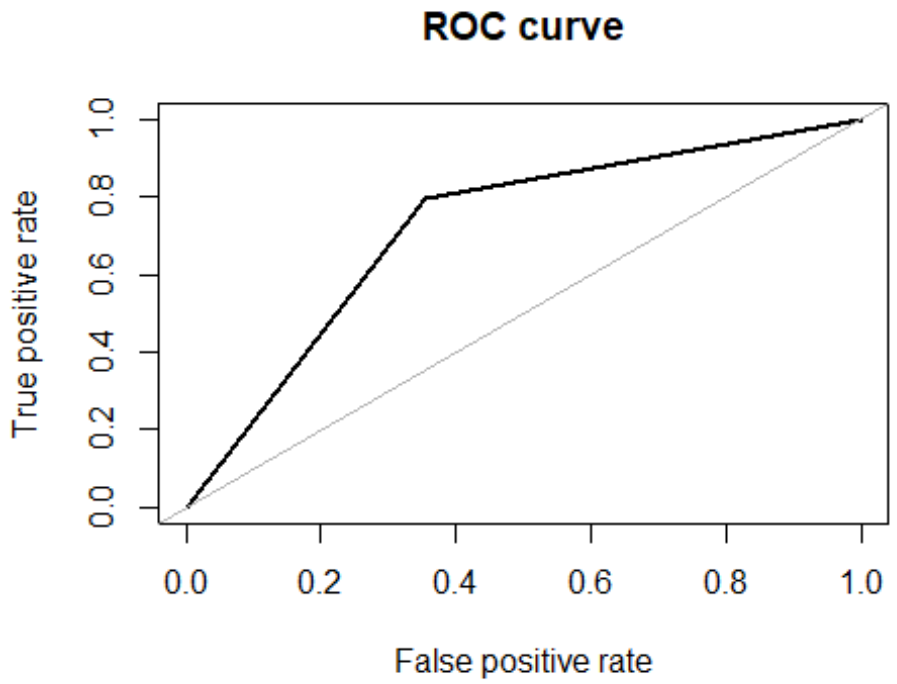

  lasso.mod=glmnet(x_train,y_train,alpha=1,lambda=grid, family = "binomial")
  lass_class <- predict(lasso.mod,s=bestlamda,newx=x_test,type="class")
  lass_pred <- rep("Attrited Customer", length(lass_class))
  lass_pred[lass_class == 1] = "Existing Customer"
  lass_acc[i] <- mean(bank[-train, ]$Attrition_Flag == lass_pred)
  lass_table <- table(lass_pred, bank[-train, ]$Attrition_Flag)
  lass_tp[i] <- lass_table[1,1]
  lass_tn[i] <- lass_table[2,2]
  lass_fp[i] <- lass_table[1,2]
  lass_fn[i] <- lass_table[2,1]
  lass_pre[i] <- lass_tp[i]/(lass_tp[i] + lass_fp[i])
  lass_rec[i] <- lass_tp[i]/(lass_tp[i] + lass_fn[i])
  lass_f1[i] <- 2*lass_pre[i]*lass_rec[i]/(lass_pre[i] + lass_rec[i])
}

avg_lass_acc <- mean(lass_acc)
avg_lass_tp <- mean(lass_tp)
avg_lass_tn <- mean(lass_tn)
avg_lass_fp <- mean(lass_fp)
avg_lass_fn <- mean(lass_fn)

avg_lass_table <- matrix(c(avg_lass_tp, avg_lass_fp, avg_lass_fn,
avg_lass_tn), byrow = TRUE, ncol = 2)
colnames(avg_lass_table) <- c("Attrited Customer", "Existing Customer")
rownames(avg_lass_table) <- c("Pred Attrited Customer", "Pred Existing
Customer")

avg_lass_pre <- mean(lass_pre)
avg_lass_rec <- mean(lass_rec)
avg_lass_f1 <- mean(lass_f1)

library(ROSE)
lass_auc <- roc.curve(bank$Attrition_Flag[-train], lass_pred)$auc
```




```
avg_lass_table
```

```
##                          Attrited Customer Existing Customer
## Pred Attrited Customer             831.0            357.75
## Pred Existing Customer             473.3           1347.95
```

```
avg_lass_acc
```

```
## [1] 0.7239037
```

```
avg_lass_pre
```

```
## [1] 0.6990934
```

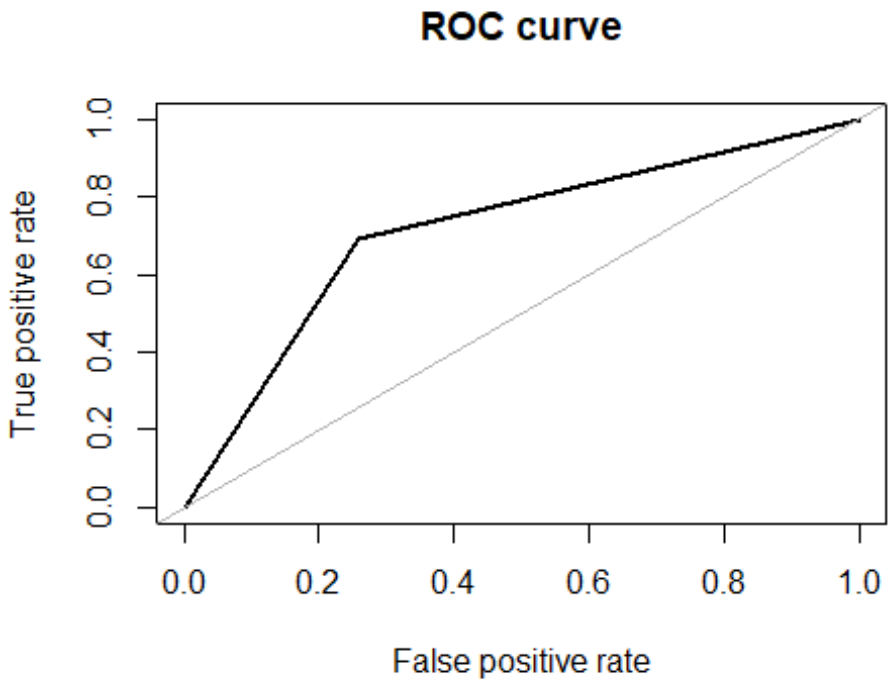
```
avg_lass_rec
```

```
## [1] 0.6372149
```

```
avg_lass_f1
```

```
## [1] 0.6666446
```

```
roc.curve(bank_pca[-train,]$bank.Attrition_Flag, pca_pred)
```




```
## Area under the curve (AUC): 0.717
```

```
fin_lasso_mod = glmnet(x_train,y_train,alpha=1,lambda=bestlamda, family =
"binomial")
fin_lasso_mod$beta
```

```
## 23 x 1 sparse Matrix of class "dgCMatrix"
##                                        s0
## Customer_Age                  -0.051744644
## GenderM                        0.314737392
## Dependent_count               -0.091631549
## Education_LevelDoctorate      -0.049221574
## Education_LevelGraduate        0.016175446
## Education_LevelHigh.School     0.018232259
## Education_LevelPost.Graduate  -0.045239141
## Education_LevelUneducated     -0.008165003
## Education_LevelUnknown         0.013204164
## Marital_StatusMarried         -0.015302314
## Marital_StatusSingle          -0.056019705
## Marital_StatusUnknown         -0.040963046
## Income_Category.40K....60K     0.276175016
## Income_Category.60K....80K     0.187655182
## Income_Category.80K....120K    0.065984896
## Income_CategoryLess.than..40K  0.304979334
## Income_CategoryUnknown         0.292750191
## Months_on_book                 0.049664875
## Total_Relationship_Count       0.466911344
```

```
## Months_Inactive_12_mon        -0.495220971
## Contacts_Count_12_mon         -0.645710740
## Credit_Limit                   0.407606273
## Avg_Utilization_Ratio          0.756662376
```

Compare with HW3

```
compare_table <- data.frame(compare_table)
compare_table

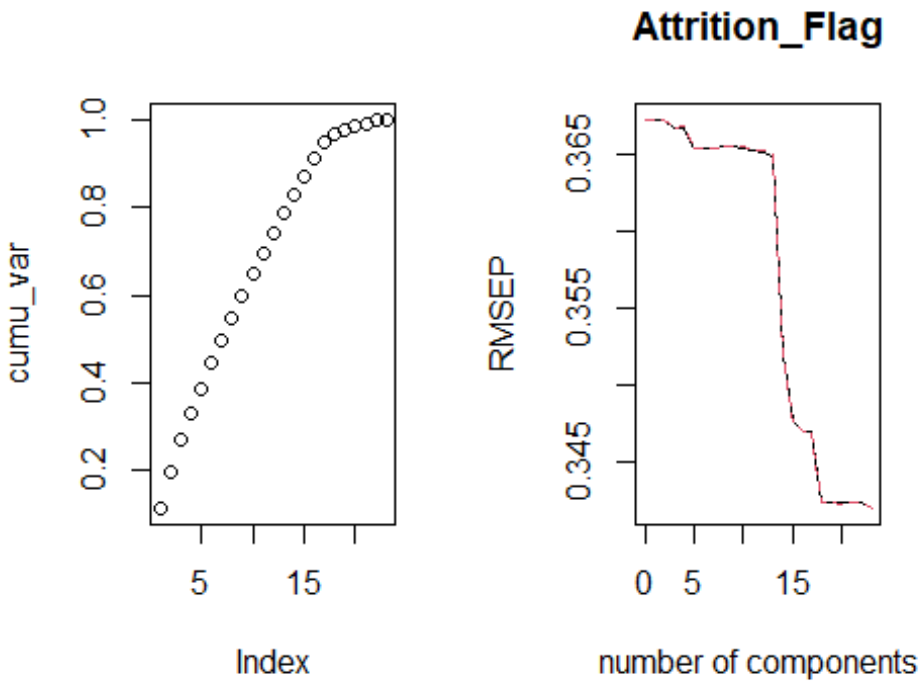
##              Model_1   Model_2   Model_3
## accuracy  0.7161296 0.7167940 0.7106645
## precision 0.6484831 0.6487609 0.6412957
## recall    0.7499732 0.7509974 0.7446498
## f1-score  0.6954680 0.6961094 0.6890582
## AUC       0.7211240 0.7268938 0.7170075

compare_table$Model_Lasso <- c(avg_lass_acc, avg_lass_pre, avg_lass_rec,
avg_lass_f1, lass_auc)
```

PCA

compare cumulative with cross-validation error

```
par(mfrow = c(1,2))
plot(cumu_var)
validationplot(pca_mod, val.type="RMSEP")
```



```
cumu_var[18]
```

```
## [1] 0.9669408
```

จากการพิจารณากราฟ จึงเลือก PCA 18

```
bank_pca <- data.frame(bank$Attrition_Flag, pca$x[,1:18])
bank_pca$bank.Attrition_Flag <- factor(bank_pca$bank.Attrition_Flag, levels = c(0,1))

library(performanceEstimation)
bank_pca <- smote(bank.Attrition_Flag~., data = bank_pca, perc.over = 3.45, perc.under = 1.75)

pca_mod <- glm(bank.Attrition_Flag~.,data = bank_pca, family = "binomial")

pca_mod$loadings
```

```
## NULL
```

ใน PC 18 นั้นมี component 3 ตัวนั่นคือ

Income_Category.80K....120K => loading(0.288)

Credit_Limit => loading(-0.776)

Avg_Utilization_Ratio => loading(-0.542)

แปลความหมายได้ว่า ลูกค้าที่มีรายได้อยู่ในระดับที่มาก
จะมีเครดิตต่อธนาคารที่มากจากกำลังในการจ่าย(ไม่เบี้ยวธนาคารเพราะไม่มีตังค์)เมื่อเครดิตของลูกค้าสูงขึ้นนั่นหมายความว่า วงเงินที่
ธนาคารปล่อยให้ก็จะมีมากขึ้นและจากการที่มีเครดิตที่มากนี้ทำให้ธนาคารยอมดึงเวลาจ่ายหนี้สินได้มากขึ้น

```
avg_pca_table
```

```
##                         Attrited Customer Existing Customer
## Pred Attrited Customer             970.75            534.25
## Pred Existing Customer             332.25           1172.75
```

```
avg_pca_acc
```

```
## [1] 0.7121262
```

```
avg_pca_pre
```

```
## [1] 0.6450166
```

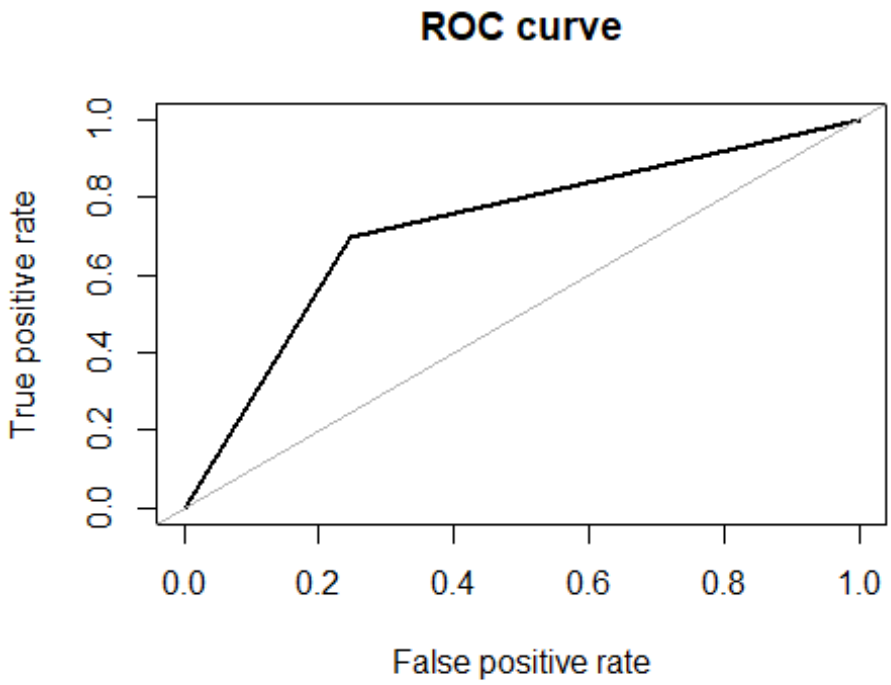
```
avg_pca_rec
```

```
## [1] 0.7451376
```

```
avg_pca_f1
```

```
## [1] 0.6914192
```

```
roc.curve(bank_pca[-train,]$bank.Attrition_Flag, pca_pred)$auc
```




```
## [1] 0.7248215
```